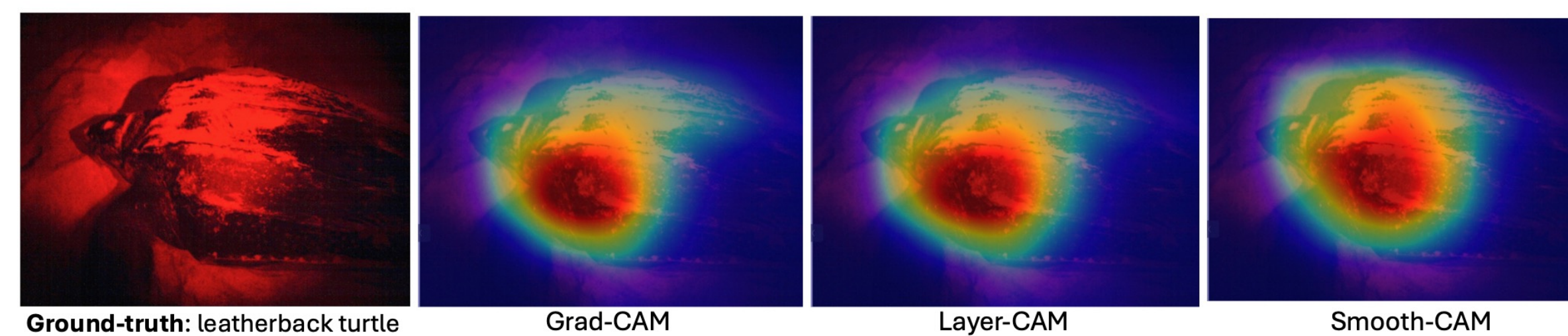
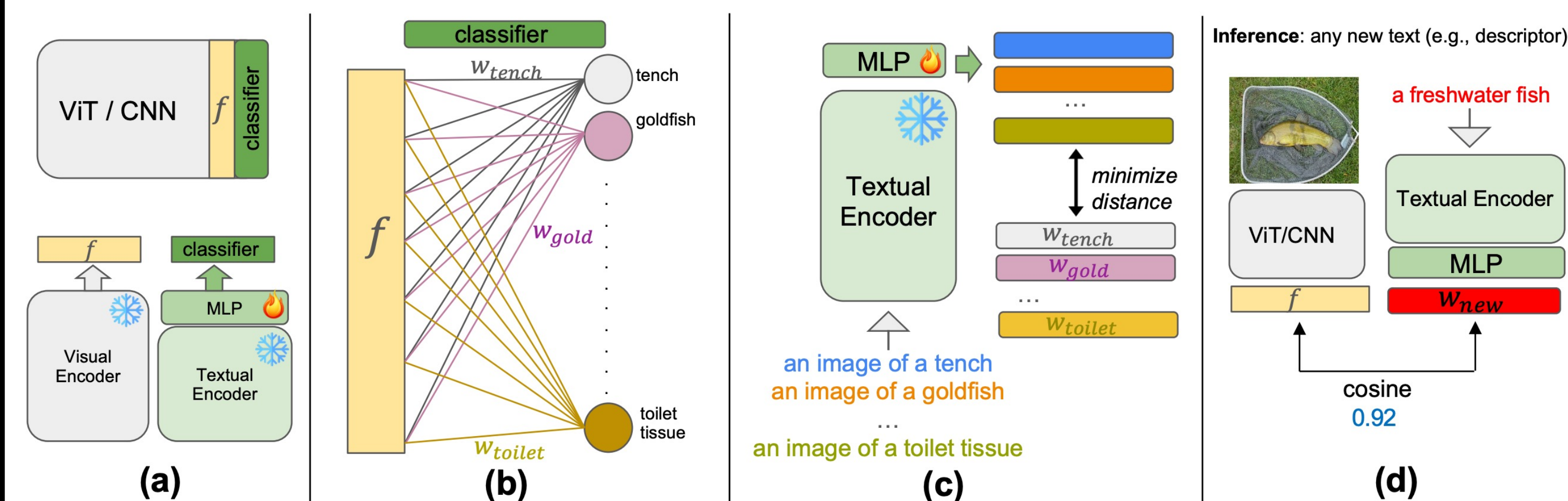


Problem Statement

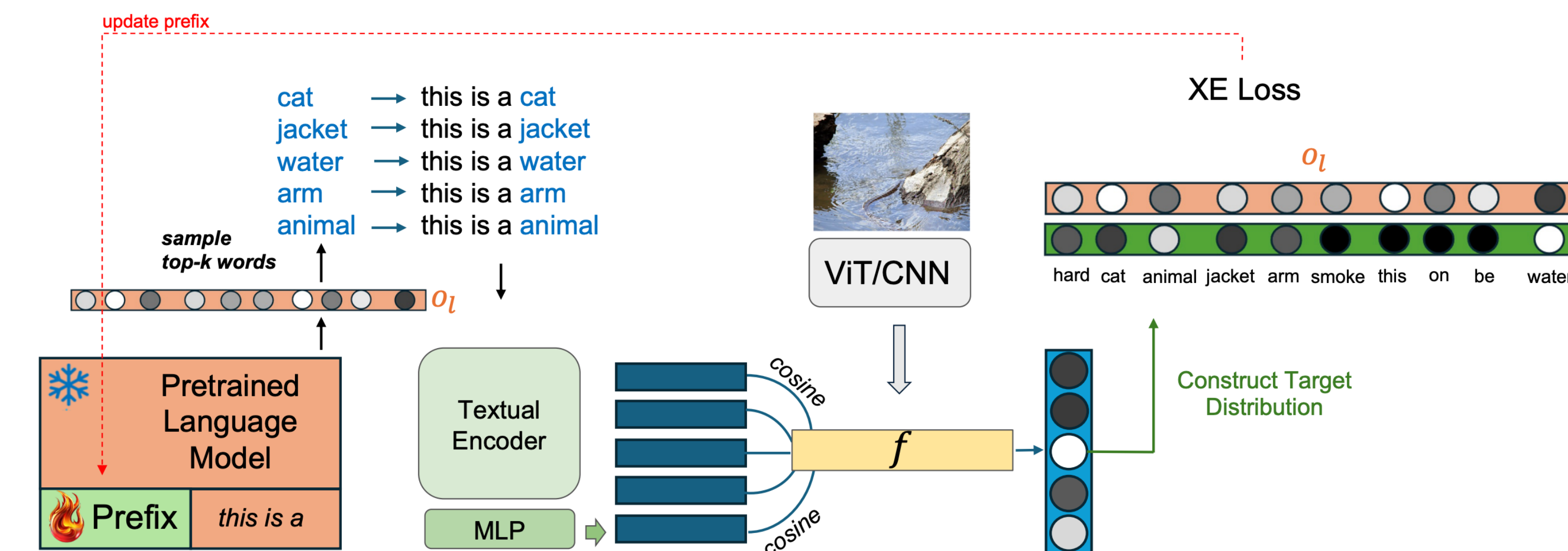
Attribution methods are not informative to understand model decisions



Learn the classifier weight space → effective learning of the text space that generalizes well to open-set text



Inference-time NLE generation: generate text which maximizes the similarity with the visual features

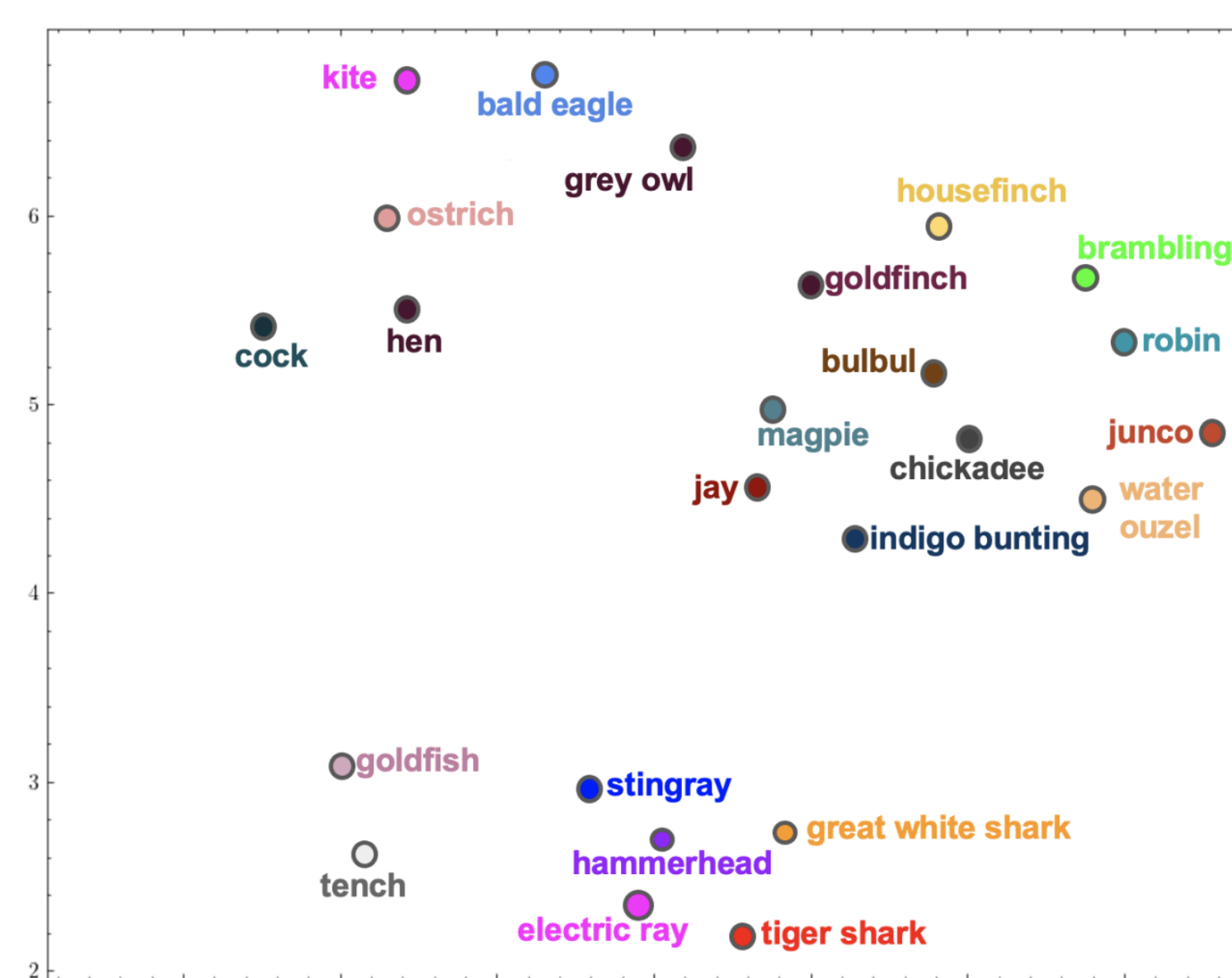


Textual Explanations solve the problem, but need annotations, rendering them **unfaithful**, reflecting the annotators reasoning process

How can we access visual models via any natural text?

Motivation

Classifier Weights of Vision models encode semantically meaningful text embeddings

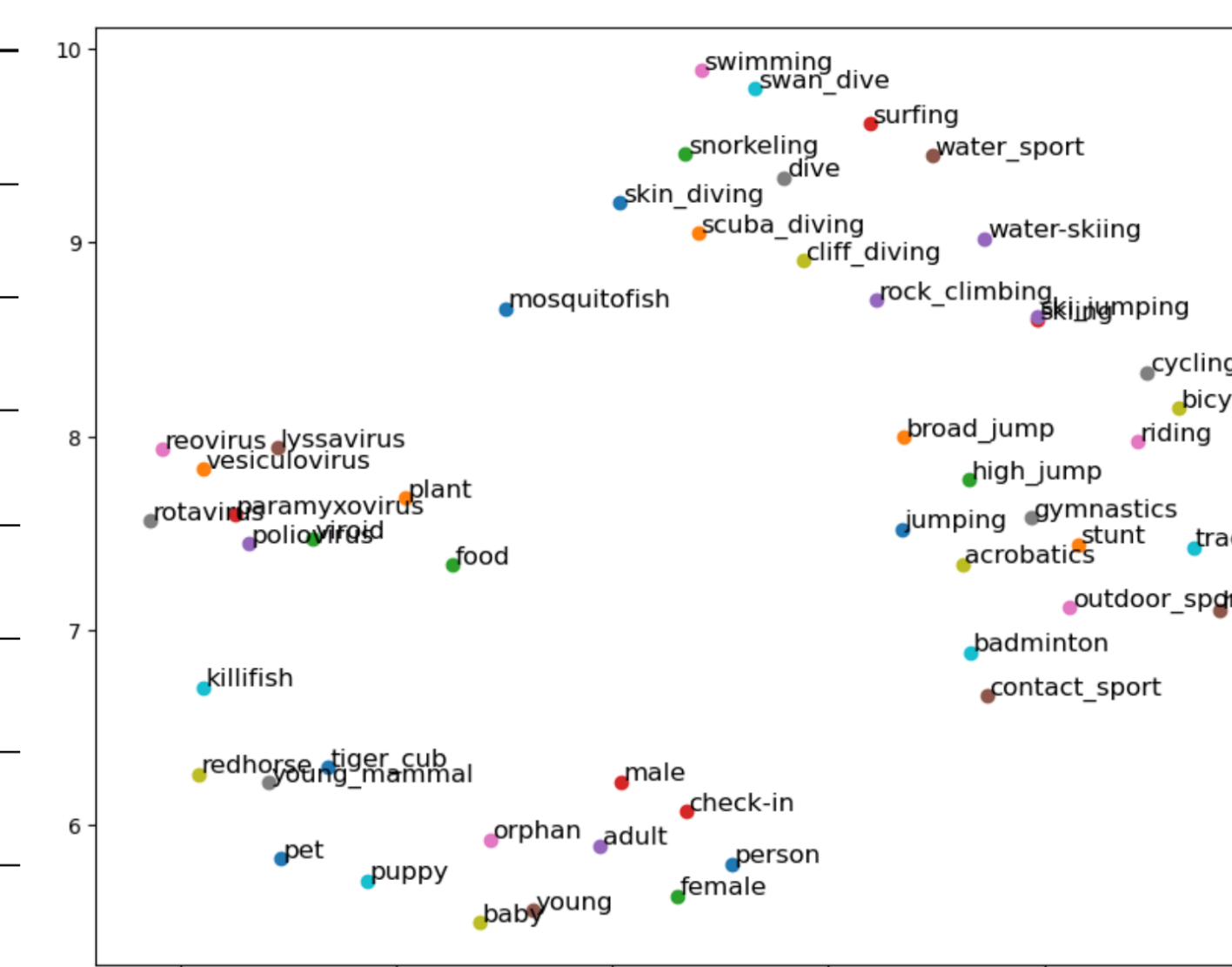


Vision Models are secretly **Vision-Language Models**

Zero-Shot Image Classification

Model	Method	Generalized Zero-Shot Setting				Zero-Shot Setting	
		Train@1	Train@5	Test@1	Test@5	Test@1	Test@5
ResNet50	ICIS	74.18	91.76	13.04	51.68	49.80	77.04
	Ours	66.77	89.34	33.28	66.20	50.88	77.62
ResNet101	ICIS	76.05	92.86	11.18	43.26	46.06	73.48
	Ours	69.42	90.19	25.92	57.80	44.36	72.96
MobileNetv3-L	ICIS	74.22	91.53	3.30	21.44	39.56	65.56
	Ours	68.26	89.84	18.90	43.42	39.32	66.00
ViT-B/16 _{swag}	ICIS	85.43	97.11	4.98	50.62	57.72	84.18
	Ours	85.47	97.35	9.18	53.84	58.20	83.72
BEiT-L/16	ICIS	87.79	98.32	5.72	49.62	56.96	78.84
	Ours	87.57	98.26	10.84	51.52	54.10	80.22
DINOv2-B	ICIS	81.30	96.08	13.96	44.70	46.60	72.14
	Ours	79.26	95.95	21.52	49.36	43.50	73.72
ConvNeXtV2-B _{pt-384}	ICIS	87.89	98.20	6.06	48.64	56.36	76.74
	Ours	87.90	98.31	9.36	50.60	56.72	77.58

Evaluation



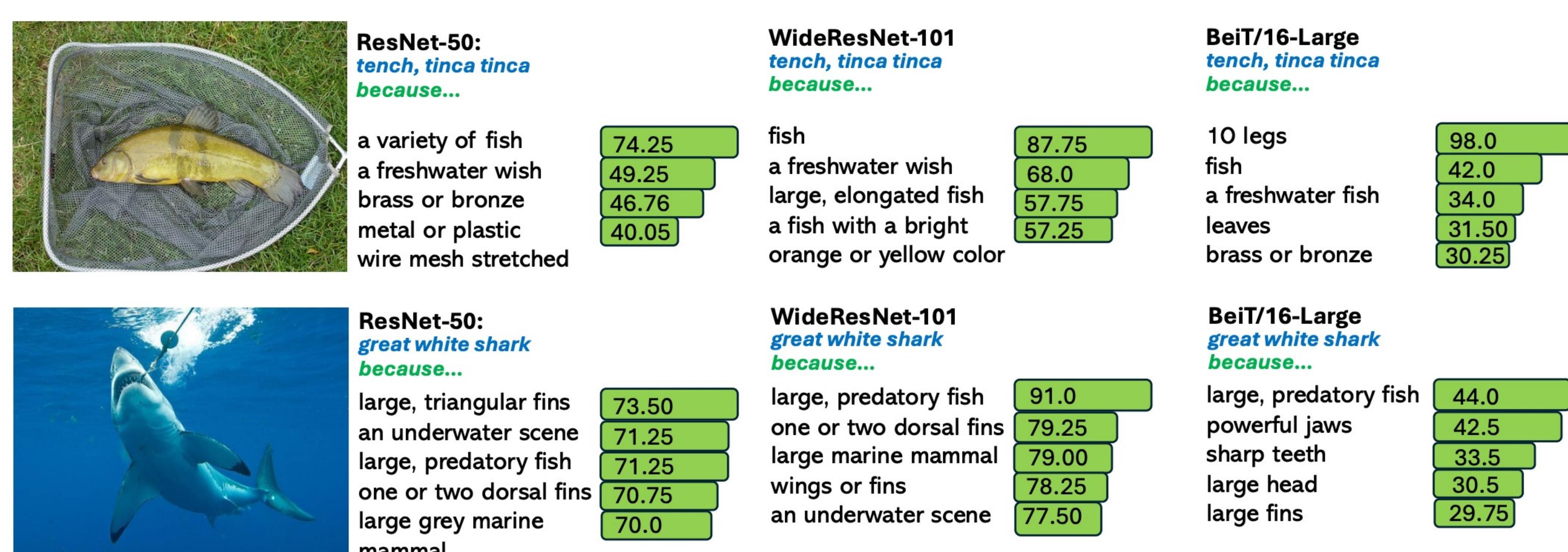
Annotation-Free Textual Explanation Evaluation

Model	Metric	NLX-GPT	NLX-GPT _r	Uni-NLX	Uni-NLX _r	Ours
ResNet50	CLIP-S ↑	28.53	28.52	28.38	28.52	30.56
	LPIPS(A) ↓	0.721	0.719	0.720	0.716	0.706
	LPIPS(S) ↓	0.624	0.623	0.620	0.620	0.614
	Cosine ↑	0.703	0.704	0.703	0.710	0.708
DenseNet161	CLIP-S ↑	28.53	28.52	28.38	28.52	30.56
	LPIPS(A) ↓	0.717	0.718	0.718	0.723	0.706
	LPIPS(S) ↓	0.620	0.620	0.620	0.624	0.616
	Cosine ↑	0.589	0.591	0.589	0.588	0.588
DINOv2-B	CLIP-S ↑	28.53	28.52	28.38	28.52	30.15
	LPIPS(A) ↓	0.719	0.716	0.717	0.716	0.712
	LPIPS(S) ↓	0.622	0.621	0.619	0.619	0.619
	Cosine ↑	0.362	0.360	0.354	0.356	0.314
ViT-B/16 _{pt}	CLIP-S ↑	28.53	28.52	28.38	28.52	30.92
	LPIPS(A) ↓	0.723	0.723	0.717	0.722	0.710
	LPIPS(S) ↓	0.624	0.626	0.619	0.624	0.621
	Cosine ↑	0.450	0.440	0.447	0.445	0.425



Beyond Textual Explanations...CLIPify ANY visual classifier in just 10 seconds with our method

Zero-Shot Concept Discovery



Classification with Concepts

Model	Baseline		+ Concepts	
	Top-1	Top-5	Top-1	Top-5
ResNet50	63.42	87.02	73.66	92.18
ResNet101	65.07	86.95	75.32	92.91
ConvNeXtV2-B	77.31	91.83	84.58	97.11
ConvNeXtV2-B _{pt}	78.93	93.32	85.96	97.87
ViT-B/32	68.66	85.76	75.31	92.34
ViT-B/16 _{swag}	77.84	93.00	84.42	97.48
ViT-L/32	69.83	86.66	76.47	92.97
ViT-L/16	72.35	88.58	79.44	94.51
Swin-S	75.40	90.66	82.45	96.34
ViT-B/16 _{pt}	76.90	92.77	82.40	96.79
BEiT-B/16	77.27	92.47	84.56	97.37
BEiT-L/16	79.90	93.58	87.06	98.22
DINOv2-B	73.49	91.29	81.99	96.37

Zero-Shot Image Captioning

Method	B@4	M	C	S
ZeroCap	2.6	11.5	14.6	5.5
ConZIC	1.3	11.5	12.8	5.2
Ours				
ResNet50	1.7	11.2	14.9	6.6
ResNet101	1.8	11.3	15.5	6.6
DINOv2-B	1.8	11.6	16.2	7.1
ConvNeXtV2-B _{pt-384}	2.0	11.5	16.9	7.2
ViT-B/16 _{pt}	1.7	11.9	17.4	7.4
BEiT-L/16	1.8	11.7	17.6	7.6

Automatic SAE feature naming

